

Homework 7

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MATH 421 — The Theory of Single Variable Calculus
Section 003
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Collaboration Statement

I went to course assistant 5:30-7pm on 10/30 namely to get a direction/high level overview on how to do problem 6 because I was high key overthinking it. All written work and final write-up are my own.

Problem 1 (*Ross Chapter 2 11.4*)

Repeat Exercise 11.2 for the sequences:

$$w_n = (-2)^n, x_n = 5^{(-1)^n}, y_n = 1 + (-1)^n, z_n = n \cos\left(\frac{n\pi}{4}\right).$$

Question 11.2 asks you to do the following for each sequence:

- (a) For each sequence, give an example of a monotone subsequence.
- (b) For each sequence, give its set of subsequential limits.
- (c) For each sequence, give its $\limsup$ and $\liminf$.
- (d) Which of the sequences converges? diverges to $+\infty$? diverges to $-\infty$?
- (e) Which of the sequences is bounded?

Discussion.

So we will analyze each sequence one by one, finding monotone subsequences (which we know must exist by 11.4 Theorem), subsequential limits, $\limsup$ and $\liminf$, convergence/divergence, and boundedness for each.

So from textbook, we know that a *monotone subsequence* is a subsequence that is either nonincreasing or nondecreasing.

A *subsequential limit* is a limit of some subsequence of the original sequence.

The $\limsup$ is the supremum (least upper bound) of the set of subsequential limits, and the $\liminf$ is the infimum (greatest lower bound) of the set of subsequential limits.

We also have a ton of helpful theorems to use from the textbook, including:

- **11.2 Theorem:** Let (s_n) be a sequence.
 - (i) If $t \in \mathbb{R}$, then $\exists$ a subsequence of (s_n) that converges to t if and only if the set $\{n \in \mathbb{N} : |s_n - t| < \epsilon\}$ is infinite for every $\epsilon > 0$.
 - (ii) If (s_n) is unbounded above, it has a subsequence with limit $+\infty$.
 - (iii) If (s_n) is unbounded below, it has a subsequence with limit $-\infty$.

In either case, the subsequence can be taken to be monotonic.

- **11.3 Theorem:** If the sequence (s_n) converges, then every subsequence of (s_n) converges to the same limit.
- **11.4 Theorem:** Every sequence has a monotone subsequence.
- **11.5 Theorem (Bolzano-Weierstrass Theorem):** Every bounded sequence has a convergent subsequence.
- **11.7 Theorem:** Let (s_n) be a sequence. There exists a monotonic subsequence whose limit is $\limsup s_n$, and there exists a monotonic subsequence whose limit is $\liminf s_n$.
- **11.8 Theorem:** Let (s_n) be a sequence in $\mathbb{R}$, let S be the set of subsequential limits of (s_n) .
 - (i) $S \neq \emptyset$
 - (ii) $\sup S = \limsup s_n$ and $\inf S = \liminf s_n$
 - (iii) $\lim s_n$ exists $\iff S$ contains exactly 1 element (namely $\lim s_n$)
- **11.9 Theorem:** Let S denote the set of subsequential limits of a sequence (s_n) . Suppose (t_n) is a sequence in $S \cap \mathbb{R}$ and that $t = \lim t_n$. Then $t \in S$.

TL;DR: The set of subsequential limits is closed.

So with all this in the back of our minds (or in front of my face because I have not yet fully baked these theorem in my head), we can use this to help find subsequential limits for each sequence.

Answers.

For $w_n = (-2)^n$, we know the sequence follows the pattern: $(-2, 4, -8, 16, -32, 64, \dots)$ whose even terms diverge to $+\infty$ and odd terms to $-\infty$ as n increases. With this in mind:

- (a) Monotone subsequence of w_n could be the even terms: $(4, 16, 64, \dots)$ which is increasing wherein the selector function could be $\sigma(k) = 2k$ for $k \in \mathbb{N}$. The other monotone subsequence could be the odd terms: $(-2, -8, -32, \dots)$ which is decreasing wherein the selector function could be $\sigma(k) = 2k - 1$ for $k \in \mathbb{N}$.

- (b) The set of subsequential limits is $\{+\infty, -\infty\}$ since we have subsequences that diverge to both $+\infty$ and $-\infty$.
- (c) $\limsup w_n = +\infty$ and $\liminf w_n = -\infty$.
- (d) Its subsequences diverge to both $+\infty$ and $-\infty$, however I do not think the sequence itself diverges to either $+\infty$ or $-\infty$ since it oscillates between large positive and negative values.
- (e) The sequence is unbounded.

For $x_n = 5^{(-1)^n}$, we see the pattern $(\frac{1}{5}, 5, \frac{1}{5}, 5, \dots)$ —odd indices give $\frac{1}{5}$ and even indices give 5. With this in mind:

- (a) A monotone subsequence of x_n could be the odd terms: $(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \dots)$, which is constant (hence nondecreasing). A selector is $\sigma(k) = 2k - 1$ for $k \in \mathbb{N}$. The other monotone subsequence is the even terms: $(5, 5, 5, \dots)$, also constant (hence nondecreasing), with selector $\sigma(k) = 2k$.
- (b) The set of subsequential limits is $\{\frac{1}{5}, 5\}$, since these two constant subsequences converge to $\frac{1}{5}$ and to 5.
- (c) $\limsup_{n \rightarrow \infty} x_n = 5$ and $\liminf_{n \rightarrow \infty} x_n = \frac{1}{5}$.
- (d) The subsequences of this sequence converge to two different limits, but the sequence itself does not seem to converge or diverge.
- (e) The sequence is *bounded* (indeed, $\frac{1}{5} \leq x_n \leq 5$ for all n).

For $y_n = 1 + (-1)^n$, the terms simply alternate $2, 0, 2, 0, \dots$ so there are two “flat” strands to exploit. With this in mind:

- (a) A monotone subsequence of y_n could be the even indices: $(2, 2, 2, \dots)$, which is constant (hence nondecreasing). A selector is $\sigma(k) = 2k$. The other monotone subsequence is the odd indices: $(0, 0, 0, \dots)$, also constant (hence nondecreasing), with selector $\sigma(k) = 2k - 1$.
- (b) The set of subsequential limits is $\{0, 2\}$, since we have constant subsequences converging to 0 and to 2.
- (c) $\limsup y_n = 2$ and $\liminf y_n = 0$.
- (d) The subsequences of this sequence converge to two different limits, but the sequence itself does not seem to converge or diverge.
- (e) The sequence is bounded (indeed, $0 \leq y_n \leq 2$ for all n).

$z_n = n \cos\left(\frac{n\pi}{4}\right)$. The cosine factor cycles with period 8 through $1, \frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2}, -1, -\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2}$, so we see large positive spikes, large negative spikes, and infinitely many zeros. With this in mind:

(a) Two clean monotone subsequences are:

- $n = 8k$: $z_{8k} = 8k$ is strictly increasing (selector $\sigma(k) = 8k$).
- $n = 8k + 4$: $z_{8k+4} = -(8k + 4)$ is strictly decreasing toward $-\infty$ (selector $\sigma(k) = 8k + 4$).

(There are also constant zero subsequences at $n = 8k + 2$ and $n = 8k + 6$.)

- (b) The set of subsequential limits is $\{-\infty, 0, +\infty\}$: along $n = 8k$ we get $+\infty$; along $n = 8k + 4$ we get $-\infty$; and along $n = 8k + 2$ (or $8k + 6$) we get 0.
- (c) $\limsup z_n = +\infty$ and $\liminf z_n = -\infty$.
- (d) The subsequences of this sequence diverge to two different limits, but the sequence itself does not seem to converge or diverge.
- (e) The sequence is unbounded (positively and negatively).

□

Problem 2 (Ross Chapter 2 11.8)

Use Definition 10.6 and Exercise 5.4 to prove $\liminf s_n = -\limsup(-s_n)$ for every sequence (s_n) .

Discussion.

To prove that $\liminf s_n = -\limsup(-s_n)$ for any sequence (s_n) , we need to connect two key ideas: the formal definitions of $\liminf$ and $\limsup$, and a useful identity from Exercise 5.4 involving infimum and supremum.

10.6 Definition:

Let (s_n) be a sequence in $\mathbb{R}$. We define:

$$\limsup s_n = \lim_{N \rightarrow \infty} \sup\{s_n : n > N\}$$

and

$$\liminf s_n = \lim_{N \rightarrow \infty} \inf\{s_n : n > N\}$$

5.4 Exercise:

Let S be a nonempty subset $\mathbb{R}$, and let $-S = -s : s \in S$. Prove $\inf S = -\sup(-S)$. *Hint: For the case $-\infty < \inf S$, simply state that this was proved in Exercise 4.9.*

Proof.

We know, from Exercise 5.4, that for any nonempty subset $S \subset \mathbb{R}$,

$$\inf S = -\sup(-S)$$

where $s \in S$ implies $-s \in -S$.

We also know that

$$\liminf s_n = \lim_{N \rightarrow \infty} \inf\{s_n : n > N\}$$

and

$$\limsup s_n = \lim_{N \rightarrow \infty} \sup\{s_n : n > N\}.$$

Using the result from Exercise 5.4, we can write

$$\inf\{s_n : n > N\} = -\sup\{-s_n : n > N\}.$$

Thus,

$$\begin{aligned} \liminf s_n &= \lim_{N \rightarrow \infty} \inf\{s_n : n > N\} \\ &= \lim_{N \rightarrow \infty} (-\sup\{-s_n : n > N\}) \\ &= -\lim_{N \rightarrow \infty} \sup\{-s_n : n > N\} \\ &= -\limsup(-s_n). \end{aligned}$$

Therefore, we have shown that

$$\liminf s_n = -\limsup(-s_n)$$

□

Problem 3 (*Ross Chapter 2 11.10*)

Let (s_n) be the sequence of numbers in Fig. 11.2 listed in the indicated order.

- (a) Find the set S of subsequential limits of (s_n) .
- (b) Determine $\limsup s_n$ and $\liminf s_n$.

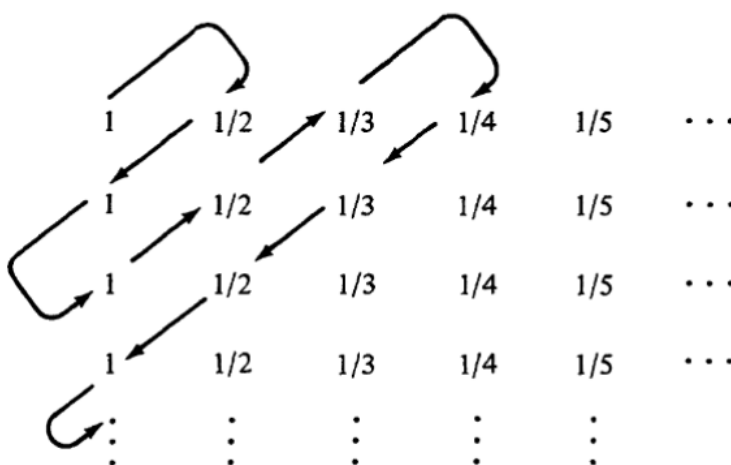


FIGURE 11.2

Discussion.

Future me — here's how I thought through this one.

So, the problem asks me to find the set of all subsequential limits of the sequence shown in the figure. The key here is remembering what a subsequential limit actually means: it is any number that some subsequence of s_n converges to.

In this particular figure, the sequence goes through the numbers $\frac{1}{n}$ infinitely often. That means each $\frac{1}{n}$ keeps reappearing infinitely many times as n runs through the natural numbers. Because of that, I can create a constant subsequence for each fixed n — that is, just pick the same $\frac{1}{n}$ value over and over again. So each $\frac{1}{n}$ is a subsequential limit.

But we also know from earlier theorems that the set of all subsequential limits is closed. So if we take all the $\frac{1}{n}$ values, their limit points must also be included in the set. Since $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$, that means 0 must also be in the set of subsequential limits.

Once we know the set S , the second part is easy: $\limsup s_n$ is just the supremum of S and $\liminf s_n$ is the infimum of S . Since S is bounded between 0 and 1, we'll get $\limsup s_n = 1$ and $\liminf s_n = 0$.

That's the entire logic chain: first find all constant subsequences, then close the set, and finally use the definitions of $\limsup$ and $\liminf$.

Proof.

(a) To find the set of subsequential limits, note that $\frac{1}{n}$ appears infinitely many times for all $n \in \mathbb{N}$.

Therefore, if we fix n , the constant subsequence consisting of $\frac{1}{n}$ for all terms converges to $\frac{1}{n}$. Hence, each $\frac{1}{n}$ is a subsequential limit, and the set of subsequential limits appears to be

$$\left\{ \frac{1}{n} : n \in \mathbb{N} \right\}.$$

However, since the set of subsequential limits must be closed, it must contain all of its limit points.

Because

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0,$$

the point 0 must also belong to S .

Thus, the set of subsequential limits is

$$S = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\} \cup \{0\}.$$

(b) From the definitions, we know that

$$\limsup s_n = \sup S \quad \text{and} \quad \liminf s_n = \inf S.$$

Since the largest element in S is 1 and the smallest is 0, it follows that

$$\limsup s_n = 1 \quad \text{and} \quad \liminf s_n = 0.$$

□

Problem 4 (*Ross Chapter 2 12.2*)

Prove $\limsup |s_n| = 0$ if and only if $\lim s_n = 0$. **Discussion.**

Future me — here is the clean way to see this. One direction is immediate: if $s_n \rightarrow 0$, then by continuity of the absolute value, $|s_n| \rightarrow 0$, so the largest tail supremum of $|s_n|$ is 0, i.e., $\limsup |s_n| = 0$. For the converse, the tail-definition of $\limsup$ says

$$\limsup |s_n| = \lim_{N \rightarrow \infty} \sup\{|s_n| : n > N\} = 0.$$

This means that for every $\varepsilon > 0$, there is N such that *all* tail terms satisfy $|s_n| < \varepsilon$ for $n > N$. That is exactly the ε - N definition of $s_n \rightarrow 0$. If I want an alternate viewpoint, I can also note that $|s_n| \geq s_n$ and $|s_n| \geq -s_n$, hence $\limsup |s_n| \geq \limsup s_n$ and $\limsup |s_n| \geq \limsup(-s_n) = -\liminf s_n$, which forces $\limsup s_n \leq 0 \leq \liminf s_n$ and therefore $\lim s_n = 0$.

Proof.

($\Rightarrow$) Suppose $\lim s_n = 0$. Then $|s_n| \rightarrow 0$ by continuity of $x \mapsto |x|$. Hence

$$\limsup |s_n| = 0.$$

($\Leftarrow$) Suppose $\limsup |s_n| = 0$. By the tail definition of $\limsup$,

$$\forall \varepsilon > 0 \exists N \text{ such that } \sup\{|s_n| : n > N\} < \varepsilon.$$

Therefore, for all $n > N$ we have $|s_n| < \varepsilon$, i.e. $-\varepsilon < s_n < \varepsilon$. Since this holds for every $\varepsilon > 0$, it follows that

$$\lim_{n \rightarrow \infty} s_n = 0.$$

This completes the proof. □

Problem 5 (*Ross Chapter 2 12.3*)

Let (s_n) and (t_n) be the following sequences that repeat in cycles of four:

$$(s_n) = (0, 1, 2, 1, 0, 1, 2, 1, 0, 1, 2, 1, 0, 1, 2, 1, 0, \dots)$$

$$(t_n) = (2, 1, 1, 0, 2, 1, 1, 0, 2, 1, 1, 0, 2, 1, 1, 0, 2, \dots)$$

Find

- | | |
|---------------------------------|---------------------------------|
| (a) $\liminf s_n + \liminf t_n$ | (e) $\limsup s_n + \limsup t_n$ |
| (b) $\liminf(s_n + t_n)$ | (f) $\liminf(s_n t_n)$ |
| (c) $\liminf s_n + \limsup t_n$ | (g) $\limsup(s_n t_n)$ |
| (d) $\limsup(s_n + t_n)$ | |

Discussion.

Future me — both (s_n) and (t_n) just cycle through $\{0, 1, 2\}$. So their subsequential limits are exactly $\{0, 1, 2\}$, giving $\liminf = 0$ and $\limsup = 2$ for each. For the sum and product, write one cycle explicitly, read off the values that reappear infinitely often, and then take $\inf / \sup$ of those sets.

Proof.

First note

$$\begin{aligned} S_s &= \{0, 1, 2\}, & \liminf s_n &= 0, & \limsup s_n &= 2, \\ S_t &= \{0, 1, 2\}, & \liminf t_n &= 0, & \limsup t_n &= 2. \end{aligned}$$

For the sum $u_n := s_n + t_n$, one period is

$$(2, 2, 3, 1),$$

so the set of subsequential limits of (u_n) is $\{1, 2, 3\}$ and hence

$$\liminf(s_n + t_n) = 1, \quad \limsup(s_n + t_n) = 3.$$

For the product $v_n := s_n t_n$, one period is

$$(0, 1, 2, 0),$$

so the set of subsequential limits of (v_n) is $\{0, 1, 2\}$ and hence

$$\liminf(s_n t_n) = 0, \quad \limsup(s_n t_n) = 2.$$

Now report each requested quantity:

- (a) $\liminf s_n + \liminf t_n = 0 + 0 = 0,$
- (b) $\liminf(s_n + t_n) = 1,$
- (c) $\liminf s_n + \limsup t_n = 0 + 2 = 2,$
- (d) $\limsup(s_n + t_n) = 3,$
- (e) $\limsup s_n + \limsup t_n = 2 + 2 = 4,$
- (f) $\liminf(s_n t_n) = 0,$
- (g) $\limsup(s_n t_n) = 2.$

□

Problem 6 (*Ross Chapter 2 12.10*)

Prove (s_n) is bounded if and only if $\limsup |s_n| < +\infty$.

Discussion.

Future me — the two directions are: (1) If (s_n) is bounded, then there is an M with $|s_n| \leq M$

for all n . Hence every tail supremum $\sup\{|s_n| : n > N\} \leq M$, so the limit of these tail suprema (i.e., $\limsup|s_n|$) is $\leq M < +\infty$. (2) Conversely, write $V_N = \sup\{|s_n| : n > N\}$. If $\limsup|s_n| < +\infty$, say $\lim_{N \rightarrow \infty} V_N = M \in \mathbb{R}$, then for large N all tail suprema are $< M + 1$, which makes the tail of (s_n) bounded by $M + 1$. Fold the first finitely many terms into the bound by taking a maximum; that gives a global bound. That's the whole idea.

Proof.

($\Rightarrow$) Suppose (s_n) is bounded. Then there exists $M \in \mathbb{R}$ such that $|s_n| \leq M$ for all n . Hence for every N ,

$$\sup\{|s_n| : n > N\} \leq M,$$

and therefore

$$\limsup_{n \rightarrow \infty} |s_n| = \lim_{N \rightarrow \infty} \sup\{|s_n| : n > N\} \leq M < +\infty.$$

($\Leftarrow$) Suppose $\limsup_{n \rightarrow \infty} |s_n| < +\infty$. Define the tail suprema

$$V_N = \sup\{|s_n| : n > N\} \quad (N \in \mathbb{N}).$$

By hypothesis, $V_N \rightarrow M$ for some finite $M \in \mathbb{R}$. Choose N_1 such that

$$|V_N - M| < 1 \quad \text{for all } N \geq N_1,$$

which implies

$$V_N < M + 1 \quad \text{for all } N \geq N_1.$$

In particular, for every $n > N_1$ we have

$$|s_n| \leq V_{N_1} < M + 1.$$

Set

$$U = \max\{|s_1|, |s_2|, \dots, |s_{N_1}|, M + 1\}.$$

Then $|s_n| \leq U$ for all n , so (s_n) is bounded.

This proves that (s_n) is bounded if and only if $\limsup|s_n| < +\infty$. □